

PH 101

Lecture 13

OUTLINE

- The Hamiltonian
- Hamilton's equations of motion

Hamiltonian formulation

The formulation of the laws of mechanics in terms of the Lagrangian, and of Lagrange's equations derived from it, presupposes that the mechanical state of a system is described by specifying its **generalized coordinates and velocities**. This is not the only possible mode of description, however. A number of advantages, especially in the study of certain general problems of mechanics, attach to a description in terms of the **generalized coordinates and momenta of the system**. The formulation arising based on **generalized coordinates and momenta of the system** is called the **Hamiltonian formulation**. The question therefore arises of the form of the equations of motion corresponding to that formulation of mechanics.

We will see how Lagrange's equations can be reformulated as a set of first order differential equations known as Hamilton's equations. **Nothing new is added to the physics and the Hamilton formulation is not superior to that of Lagrange when it comes to problem solving.** The value of Hamilton's supremely elegant formulation lies in providing a foundation for theoretical extensions both within and outside classical mechanics. Within classical mechanics, it is the basis for most further developments such as the Hamilton-Jacobi theory and chaos. Elsewhere, Hamiltonian mechanics provides the best route to statistical mechanics, and the notion of the Hamiltonian is at the heart of quantum mechanics.

SYSTEMS OF FIRST ORDER ODES

The standard form for a system of first order ODEs in the n unknown functions $x_1(t), x_2(t), \dots, x_n(t)$ is

$$\begin{aligned}\dot{x}_1 &= F_1(x_1, x_2, \dots, x_n, t), \\ \dot{x}_2 &= F_2(x_1, x_2, \dots, x_n, t), \\ &\vdots \\ \dot{x}_n &= F_n(x_1, x_2, \dots, x_n, t),\end{aligned}$$

where $F_1, F_2, \dots, F_n$ are given functions of the variables $x_1, x_2, \dots, x_n, t$. This can also be written in the compact vector form

$$\dot{\mathbf{x}} = \mathbf{F}(\mathbf{x}, t),$$

where $\mathbf{x}$ and $\mathbf{F}$ are the n -dimensional vectors $\mathbf{x} = (x_1, x_2, \dots, x_n)$ and $\mathbf{F} = (F_1, F_2, \dots, F_n)$. If the value of $\mathbf{x}$ is given when $t = t_0$, the equations $\dot{\mathbf{x}} = \mathbf{F}(\mathbf{x}, t)$ determine the unknowns $\mathbf{x}(t)$ at all subsequent times.

Example

Consider the orbit equations for a particle of mass $\mathbf{m}$ attracted by the gravitation of a mass $\mathbf{M}$ fixed at the origin $\mathbf{O}$. In terms of polar coordinates centred on $\mathbf{O}$, the Lagrangian is

$$L = \frac{1}{2}m \left(\dot{r}^2 + r^2 \dot{\theta}^2 \right) + \frac{mMG}{r}.$$

and the corresponding Lagrange equations are

$$\ddot{r} - r\dot{\theta}^2 = -\frac{MG}{r^2}, \qquad 2r\dot{r}\dot{\theta} + r^2\ddot{\theta} = 0,$$

a pair of second order ODEs in the unknowns $\{r(t), \theta(t)\}$. If we now introduce the new variables v_r and v_θ , defined by

$$v_r = \dot{r}, \qquad v_\theta = \dot{\theta},$$

$$\ddot{r} - r\dot{\theta}^2 = -\frac{MG}{r^2}, \quad 2r\dot{r}\dot{\theta} + r^2\ddot{\theta} = 0.$$

The two second order Lagrange equations can be converted into

$$\begin{aligned} \dot{r} &= v_r, & \dot{v}_r &= rv_\theta^2 - MG/r^2, \\ \dot{\theta} &= v_\theta, & \dot{v}_\theta &= -2v_rv_\theta/r, \end{aligned}$$

an equivalent system of *four first order* ODEs in the four unknowns $\{r, \theta, v_r, v_\theta\}$.

Hamilton form

In the above orbit equations example, we have performed the conversion to first order equations by introducing the **coordinate velocities** as new variables. This is not the only choice however. When transforming a system of Lagrange equations to a first order system, we could instead take the **conjugate momenta**, defined by

$$p_j = \frac{\partial L}{\partial \dot{q}_j}, \quad (1)$$

as the new variables. This seems quite attractive since the Lagrange equations already have the form

$$\dot{p}_j = \frac{\partial L}{\partial q_j} \quad (1 \leq j \leq n).$$

The downside is that the right sides $\frac{\partial L}{\partial q_j}$ are functions of $q, \dot{q}, t$ and must be transformed to functions of q, p, t . This is achieved by inverting the equations (1) to express the $\dot{q}$ as functions of q, p, t .

$$L = \frac{1}{2}m \left(\dot{r}^2 + r^2 \dot{\theta}^2 \right) + \frac{mMG}{r} \quad : \text{Lagrangian for the orbit problem}$$

For the Lagrangian in the orbit problem, the conjugate momenta are

$$p_r = \frac{\partial L}{\partial \dot{r}} = m\dot{r}, \quad p_\theta = \frac{\partial L}{\partial \dot{\theta}} = mr^2\dot{\theta},$$

and these equations are easily inverted to give

$$\dot{r} = \frac{p_r}{m}, \quad \dot{\theta} = \frac{p_\theta}{mr^2}.$$

The *two second order* Lagrange equations for the orbit problem are therefore equivalent to the system of *four first order* ODEs

$$\begin{aligned}\dot{r} &= \frac{p_r}{m}, & \dot{p}_r &= \frac{p_\theta^2}{mr^3} - \frac{mMG}{r^2}, \\ \dot{\theta} &= \frac{p_\theta}{mr^2}, & \dot{p}_\theta &= 0,\end{aligned}$$

in the four unknowns $\{r, \theta, p_r, p_\theta\}$. This is the **Hamilton form** of Lagrange's equations for the orbit problem.

All of you are probably wondering what is the point of converting Lagrange's equations into a system of first order ODEs. For the purpose of finding solutions to particular problems, like the orbit problem above, there is no point. Indeed, the new system of first order equations may be harder to solve than the original second order equations. The real interest lies in the structure of the general theory. When Lagrange's equations are expressed in Hamilton form in a general manner, the result is a system of first order equations of great simplicity and elegance, now known as Hamilton's equations. **The Hamiltonian function, which appears in Hamilton's equations, is at the heart of quantum mechanics.**

The Hamiltonian function

Consider the Lagrangian function for a system as a function of just one generalized coordinate and generalized velocity.

$$L = L(q, \dot{q})$$

The total differential of the Lagrangian as a function of coordinates and velocities is

$$dL = \frac{\partial L}{\partial q} dq + \frac{\partial L}{\partial \dot{q}} d\dot{q} \rightarrow (1)$$

Because of Lagrange's equation,

$$\frac{\partial L}{\partial q} = \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}} \right)$$

and $p = \frac{\partial L}{\partial \dot{q}}$, we can rewrite (1) as:

$$dL = \dot{p} dq + p d\dot{q} \rightarrow (2)$$

Just like,

$$d(xy) = x dy + y dx$$

we can write:

$$d(\dot{q} p) = \dot{q} dp + p d\dot{q} \rightarrow (3)$$

Using (3), the second term in (2) can be re-written as:

$$p d\dot{q} = d(\dot{q} p) - \dot{q} dp$$

Thus, we obtain from (2) :

$$dL = \dot{p} dq + d(\dot{q} p) - \dot{q} dp$$

$$\Rightarrow \boxed{d(\dot{q} p - L) = -\dot{p} dq + \dot{q} dp} \rightarrow (4)$$

The argument of the differential on
Left hand side of (4), i.e.

$\dot{q} p - L$ is the energy 'E', discussed
in Lecture 12, written in terms of
coordinates and momentum.

This function, $\dot{q}p - L$ is called the Hamilton's function or Hamiltonian, H , of the system.

$$H = \dot{q}p - L$$

We can extend our treatment to a set of n generalized coordinates and momenta;

$$H(q, p, t) = \sum_i \dot{q}_i p_i - L$$

Examples

Example (Harmonic oscillator): Consider a 1-D harmonic oscillator described by the Lagrangian $L = m\dot{x}^2/2 - kx^2/2$. The conjugate momentum is $p \equiv \partial L/\partial \dot{x} = m\dot{x}$, which yields $\dot{x} = p/m$. So the Hamiltonian is

$$\begin{aligned} H = p\dot{x} - L &= p\dot{x} - \left(\frac{m}{2}\dot{x}^2 - \frac{k}{2}x^2 \right) \\ &= p \left(\frac{p}{m} \right) - \frac{m}{2} \left(\frac{p}{m} \right)^2 + \frac{k}{2}x^2 \\ &= \frac{p^2}{2m} + \frac{kx^2}{2}, \end{aligned}$$

where we now have H in terms of x and p , with no $\dot{x}$'s. H is simply the energy, expressed in terms of x and p .

Example (Central force): Consider a 2-D central-force setup described by the Lagrangian $L = m(\dot{r}^2 + r^2\dot{\theta}^2)/2 - V(r)$. The two conjugate momenta are

$$\begin{aligned}p_r &\equiv \frac{\partial L}{\partial \dot{r}} = m\dot{r} \implies \dot{r} = \frac{p_r}{m}, \\p_\theta &\equiv \frac{\partial L}{\partial \dot{\theta}} = mr^2\dot{\theta} \implies \dot{\theta} = \frac{p_\theta}{mr^2}.\end{aligned}$$

So the Hamiltonian is

$$\begin{aligned}H &= \left(\sum p_i \dot{q}_i\right) - L \\&= p_r \dot{r} + p_\theta \dot{\theta} - \left(\frac{m}{2}(\dot{r}^2 + r^2\dot{\theta}^2) - V(r)\right) \\&= p_r \left(\frac{p_r}{m}\right) + p_\theta \left(\frac{p_\theta}{mr^2}\right) - \frac{m}{2} \left(\frac{p_r}{m}\right)^2 - \frac{mr^2}{2} \left(\frac{p_\theta}{mr^2}\right)^2 + V(r) \\&= \frac{p_r^2}{2m} + \frac{p_\theta^2}{2mr^2} + V(r).\end{aligned}$$

Hamilton's Equations of Motion

The Hamiltonian, H , is always considered as a function of the (q_i, p_i, t) set whereas the Lagrangian, L , is a function of $(q_i, \dot{q}_i, t)$

$$H = H(q_i, p_i, t)$$

$$L = L(q_i, \dot{q}_i, t)$$

Now, the total differential of H is

$$dH = \sum_i \left(\frac{\partial H}{\partial q_i} dq_i + \frac{\partial H}{\partial p_i} dp_i \right) + \frac{\partial H}{\partial t} dt \rightarrow (1)$$

Again, we know:

$$H = \sum_i \dot{q}_i p_i - L$$

Thus,

$$dH = \sum_i \left(\dot{q}_i dp_i + p_i d\dot{q}_i \right) - dL$$

$$dL = \sum_i \left(\frac{\partial L}{\partial q_i} dq_i + \frac{\partial L}{\partial \dot{q}_i} d\dot{q}_i \right) + \frac{\partial L}{\partial t} dt$$

Hence,

$$dH = \sum_i \left(\dot{q}_i dp_i + p_i d\dot{q}_i - \frac{\partial L}{\partial q_i} dq_i - \frac{\partial L}{\partial \dot{q}_i} d\dot{q}_i \right) - \frac{\partial L}{\partial t} dt$$

$$= \sum_i \left(\dot{q}_i dp_i + p_i d\dot{q}_i - \dot{p}_i dq_i - p_i d\dot{q}_i \right) - \frac{\partial L}{\partial t} dt$$

$$\Rightarrow dH = \sum_i \left(\dot{q}_i dp_i - \dot{p}_i dq_i \right) - \frac{\partial L}{\partial t} dt \rightarrow (2)$$

Comparing (1) and (2) we obtain:

$$\begin{aligned}\dot{q}_i &= \frac{\partial H}{\partial p_i} \\ \dot{p}_i &= -\frac{\partial H}{\partial q_i}\end{aligned}$$

→ (3) $\left\{ \begin{array}{l} \text{Hamilton's} \\ \text{equations of} \\ \text{motion, also} \end{array} \right.$
known as
canonical equations
of motion.

and $-\frac{\partial L}{\partial t} = \frac{\partial H}{\partial t}$

From:

$$dH = \sum_i \left(\frac{\partial H}{\partial q_i} dq_i + \frac{\partial H}{\partial p_i} dp_i \right) + \frac{\partial H}{\partial t} dt$$

we can obtain:

$$\begin{aligned} \frac{dH}{dt} &= \sum_i \left(\frac{\partial H}{\partial q_i} \dot{q}_i + \frac{\partial H}{\partial p_i} \dot{p}_i \right) + \frac{\partial H}{\partial t} \\ &= \sum_i \left(\frac{\partial H}{\partial q_i} \frac{\partial H}{\partial p_i} - \frac{\partial H}{\partial p_i} \frac{\partial H}{\partial q_i} \right) + \frac{\partial H}{\partial t} \\ &= \frac{\partial H}{\partial t} \end{aligned}$$

Clearly, if H does not explicitly contain the time 't', then $\frac{\partial H}{\partial t} = 0$

$$\Rightarrow \frac{dH}{dt} = 0$$

$\Rightarrow$ The Hamiltonian, H , is a conserved quantity.

If the potential energy is velocity independent, it can be shown that the Hamiltonian equals the total energy:

$$H = T + V$$

And, if $\frac{\partial H}{\partial t} = 0$, $H = E = \text{constant.}$

To Show that: $H = T + V$

We know:

$$H = \sum_i \dot{q}_i \frac{\partial L}{\partial \dot{q}_i} - L$$

$$L = T - V(q_i)$$

Then,

$$H = \sum_i \dot{q}_i \frac{\partial T}{\partial \dot{q}_i} - (T - V)$$

According to Euler's theorem, if the function $f(y_i)$ is a homogeneous function of the y_i that is of degree n , then

$$\sum_i y_i \frac{\partial f}{\partial y_i} = n f$$

Now, T is a quadratic function of $\dot{q}_i$, i.e. of degree 2. So,

$$\sum_i \dot{q}_i \frac{\partial T}{\partial \dot{q}_i} = 2T$$

Using this result:

$$\begin{aligned} H &= 2T - (T - V) \\ &= T + V \end{aligned}$$

